DO NOW, AO1 REVISION

Radioactive decay and nuclear radiation

Atomic structure
Paper 1, Combined and Separate
26 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State which part of an atom nuclear radiation is emitted from.**	The nucleus	☐
2	**Explain why unstable nuclei emit radiation.**	To become more stable.	☐
3	**What is radioactive decay?**	The random process in which an unstable nucleus emits radiation.	☐
4 ★	**Explain why radioactive decay is described as random.**	It is impossible to predict which nucleus will decay next, or when.	☐
5 ★	**Define the activity of a radioactive source.**	The rate at which its unstable nuclei decay.	☐
6 ★	**State the standard unit of activity.**	Becquerel (Bq)	☐
7	**What is count-rate?**	The number of decays recorded each second by a detector.	☐
8 ★	**Explain why the count-rate is always less than the activity.**	The detector picks up only some of the radiation emitted.	☐
9	**Name the four types of nuclear radiation.**	Alpha particles, beta particles, gamma rays and neutrons	☐
10	**Name three types of ionising nuclear radiation.**	Alpha, beta and gamma	☐
11 ★	**State what an alpha particle is made of.**	Two protons and two neutrons (a helium nucleus)	☐
12 ★	**State what a beta particle is.**	A high-speed electron from the nucleus	☐
13 ★	**State what a gamma ray is.**	Electromagnetic radiation from the nucleus	☐

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	QUESTION	ANSWER	✓
14	**A nucleus contains no electrons. Describe how it can emit a beta particle.**	A neutron turns into a proton and an electron; the electron is emitted.	☐
15 ★	**List alpha, beta and gamma radiation from most to least ionising.**	Alpha, beta, gamma	☐
16 ★	**List alpha, beta and gamma radiation from most to least penetrating.**	Gamma, beta, alpha	☐
17 ★	**State the range of alpha radiation in air.**	A few centimetres	☐
18	**State the range of beta radiation in air.**	About a metre	☐
19	**State the range of gamma radiation in air.**	A very long distance (it is hardly absorbed by air)	☐
20	**Name the type of radiation stopped by a sheet of paper.**	Alpha	☐
21 ★	**Name the two types of radiation stopped by a few millimetres of aluminium.**	Alpha and beta	☐
22 ★	**State what is needed to absorb most gamma radiation.**	Several centimetres of lead	☐
23	**Name the type of radiation stopped by skin.**	Alpha	☐
24	**Name a detector of nuclear radiation.**	A Geiger–Müller tube	☐
25	**State the relative charges of alpha, beta and gamma radiation.**	Alpha: +2 Beta: −1 Gamma: 0	☐
26	**What is ionisation?**	An atom gaining or losing electrons to become an ion.	☐

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